

Yudong Gao

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Research Interests

AI Security · LLM & Agent Security · Embodied AI Security · Trustworthy Machine Learning

Education

Hong Kong University of Science and Technology Sep 2025 – Present

Ph.D. in Computer Science and Engineering (CSE) Advisor: Prof. Shuai Wang

- GPA: 3.780/4.0.

China University of Petroleum (East China)

Sep 2019 – Jun 2025

B.Eng.–M.Eng. Combined Program Advisor: Prof. Honglong Chen

- College of Control Science and Engineering. GPA: 90.12/100 (Rank: 1/25).

Research Experience

Research Assistant

May 2023 – May 2024

Advisor: Prof. Huajie Shao College of William & Mary, USA

- Designed an energy-based backdoor defense requiring no task-specific samples (**ICML 2024**).

Publications

- [1] **PathMark: Protecting Intellectual Property of Mixture-of-Expert LLMs via Path Watermarks**
Y. Gao, Q. Wang, Y. Yuan, R. Huang, L. Chen, Z. Ji, S. Wang. *ACM Conference on Computer and Communications Security (CCS)*, 2026. (accepted) (**CCF-A**)
- [2] **A Triple Stealthy Backdoor: Hidden in Spatial, Frequency, and Feature Domains**
Y. Gao, H. Chen, P. Sun, J. Li, Y. Yin, Z. Wang, W. Liu. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2025. (**CCF-A**)
- [3] **A Dual Stealthy Backdoor: From Both Spatial and Frequency Perspectives**
Y. Gao, H. Chen, P. Sun, J. Li, A. Zhang, Z. Wang, W. Liu. *AAAI Conference on Artificial Intelligence (AAAI)*, 2024. (**CCF-A**)
- [4] **Energy-Based Backdoor Defense Without Task-Specific Samples and Model Retraining**
Y. Gao, H. Chen, P. Sun, Z. Li, J. Li, H. Shao. *International Conference on Machine Learning (ICML)*, 2024. (**CCF-A**)
- [5] **Investigating the Backdoor on DNNs Based on Recolorization and Reconstruction**
H. Chen, Y. Gao, A. Zhang, P. Sun, W. Liu, X. Wang. *IEEE Transactions on Information Forensics and Security (TIFS)*, 2024. (**CCF-A**)
- [6] **One Cognitive Loop Is Enough: SODA Unlocks Pure-Text Spatial Reasoning in Large Language Models**
Y. Gao (corresponding author). *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. (accepted) (**CCF-A**)
- [7] **Frequency Is Devious: Deep Stealthy Backdoor in Artificial Intelligence Based Consumer IoT**
Y. Gao, H. Chen, J. Li, X. Wang, A. Alharbi, A. Tolba, F. Xia. *IEEE Transactions on Consumer Electronics (TCE)*, 2024.
- [8] **SkillReducer: Optimizing LLM Agent Skills for Token Efficiency**
Y. Gao, Z. Li, Y. Yuan, Z. Ji, P. Ma, S. Wang. *arXiv preprint*, 2026.
- [9] **Lightweight Privacy Protection Against Channel-Level Gradient Leakage Attack in Federated Learning**
Z. Li, H. Chen, Y. Gao, H. Xue. *International Conference on Information and Communication Technology Convergence (ICTC)*, 2024. **Best Paper Award**

Honors & Awards

- **National Scholarship** (Top 1%, ×3), Ministry of Education
- **Outstanding Undergraduate Thesis** 2025
- **Best Paper Award, ICTC** 2024
- **First Prize, Chinese Mathematics Competitions (CMC)** 2020

Academic Service

Journal Reviewer: IEEE TIFS, IEEE TDSC, IEEE TNNLS, Pattern Recognition, ESWA, EAAI

Conference Reviewer: NeurIPS, ACM MM

Sub-reviewer: ACM CCS, USENIX Security, FSE